

ARI CHOLAKIAN

Engineering Physics

Burnaby, British Columbia, Canada

ari.cholakian@gmail.com ♦ 17788294303 ♦ aricholakian.github.io ♦ LinkedIn: ari-mesrob-cholakian

EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Applied Science in Engineering Physics, 3.70 GPA

Sep 2023 – May 2028

- **Courses:** Machine Design, Software Construction, Signals & Systems, Fluid Mechanics II, Machine Learning, Quantum Mechanics
- **Activities:** 40-hour Machining Certification, Engineering Physics Student Council, Clubroom Manager

TECHNICAL EXPERIENCE

AME Consulting Group

Vancouver, BC

Mechanical Engineering Coop

Jan 2025 – May 2025

- Prepared and issued nearly 30 mechanical drawing sets in AutoCAD and Revit 3D for a variety of projects, including data centres, biotech labs, office renovations, multi-unit residential projects, and District Energy connections
- Designed energy-efficient and cost-effective HVAC and plumbing systems according to standards and manufacturer guidelines
- Ran flow, energy, and stress simulations using internal and industry-standard tools like AutoCAD MEP and CAEPIPE

UBC Orbit - Satellite Design Team

Captain and Lead Systems Engineer

Sep 2025 – Present

- Oversaw the high level system architecture, design, integration, and verification of the **ALEASAT** project, a student-led disaster recovery CubeSat mission at UBC with the European Space Agency (ESA)
- Designed budgets and enabled the integration of the satellite's many subsystems like power, on-board software, and communications
- Supported FlatSats, vibration, thermal and EMC testing campaigns to verify functionality
- Led the **team of 70** undergraduate and graduate engineering students to meet mission objectives and worked directly with ESA, ITU, IARU, ISED, Global Affairs Canada and Radio Amateurs to ensure regulatory compliance

Radio Frequency Communications and Structure Designer

Sep 2023 – May 2025

- Designed the Antenna Deployer Module timer PCB using Altium, followed by fabrication and SMT reflow assembly
- Designed a space-qualified burn-wire deployer mechanism and carried out environmental testing in ESA CubeSat facilities, adhering to Design for Manufacture practices and ECSS guidelines
- Simulated antenna properties and radiation patterns in Ansys Electromagnetic Suite to optimize for LEO omnidirectionality

PROJECTS

"Pet-Saving" Robot

Engineering Physics "Robot Summer" Competition

Aug 2025

- Won **1st place** with an autonomous robot that used LIDAR and IMUs to collect and move plushies, built in just **6 weeks**
- Rapidly manufactured an all-metal chassis and 3 DOF robot arm with 3D printed and manually machined parts on a lathe and mill
- Supported team in software architectural design and electrical hardware integration by maintaining OnShape/Solidworks 3D models

RESEARCH AND PUBLICATIONS

A Dynamic Channel Emulator for CubeSat System Test and Integration

Published in 2024 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)

Sep 2024

Using a Dynamic Channel Emulator for CubeSat GNSS Receiver Test and Integration

Published in 2025 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)

May 2025

SKILLS

- **Mechanical:** CAD/CAM, Solidworks, OnShape, AutoCAD, Revit, FEA, 3D Printing, Machining (lathe, mill, waterjet)
- **Electrical:** Circuit design (Altium), Ansys HFSS, Through hole and SMT soldering, Amateur Radio Operator, VNA
- **Software:** Python, Java/C#, git, MATLAB, Excel, Typst, VSCode